

REFNO: 075238

Roll No. _____

B.A./B.Sc. (Hons.) SEMESTER VI EXAMINATION 2022-23
MATHEMATICS

MTB - 603 : Vector and Tensor Analysis

Time : Three hours

Max. Marks : 70

(WRITE YOUR ROLL NO. AT THE TOP IMMEDIATELY ON THE RECEIPT OF THIS QUESTION PAPER)

Note: Answer **FIVE** questions, including **Question 1**, which is compulsory. Write the Number of a Question, while answering it.

1. (a) If $\phi(x, y, z) = xyz^2$ and $\mathbf{A} = xz\mathbf{i} + y^2\mathbf{j} + xzy\mathbf{k}$, obtain $\frac{\partial^2}{\partial x \partial y}(\phi\mathbf{A})$. (2)

(b) Given $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$, show that $\nabla(\|\mathbf{r}\|^n) = n\|\mathbf{r}\|^{n-2}\mathbf{r}$. (2)

(c) Give definition of divergence $\nabla \cdot \mathbf{V}$ of a smooth vector field

$$\mathbf{V}(x, y, z) = V_1(x, y, z)\mathbf{i} + V_2(x, y, z)\mathbf{j} + V_3(x, y, z)\mathbf{k}.$$

Obtain $\nabla \cdot \mathbf{V}$ for $\mathbf{V} = yz\mathbf{i} + xz\mathbf{j} + xy\mathbf{k}$. (1+1)

(d) Give definition of curl $\nabla \times \mathbf{V}$ of a smooth vector field

$$\mathbf{V}(x, y, z) = V_1(x, y, z)\mathbf{i} + V_2(x, y, z)\mathbf{j} + V_3(x, y, z)\mathbf{k}.$$

Obtain $\nabla \times \mathbf{V}$ for $\mathbf{V} = z\mathbf{i} + y\mathbf{j} + zx\mathbf{k}$. (1+1)

(e) Given $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$, prove that $\nabla^2 \left(\frac{1}{\|\mathbf{r}\|} \right) = 0$. (2)

(f) Give definitions of conservative and irrotational vector fields. (1+1)

(g) Give the statements of Gauss' divergence theorem and Green's theorem. (2)

(h) Define an orthogonal curvilinear coordinate system in $\mathbb{R}^3$. (2)

(i) Write down coordinate surfaces and coordinate curves for spherical coordinate system in $\mathbb{R}^3$. (2)

(j) Obtain element of arc length for cylindrical coordinate system in $\mathbb{R}^3$. (2)

(k) Give definition of contravariant and covariant vectors. (2)

2. (a) Prove that $\nabla \times (\nabla\phi) = 0$. (6)

(b) Prove that $\nabla \cdot (\nabla \times \mathbf{V}) = 0$. (6)

3. (a) If $\mathbf{V} = xy^2\mathbf{i} + yx^2\mathbf{j} + xz^2\mathbf{k}$, then evaluate $\int_{(0,0,0)}^{(1,1,1)} \mathbf{V} \cdot d\mathbf{r}$ along the path $x = t, y = t^2, z = t^3$. (6)

(b) If $\int_C \mathbf{V} \cdot d\mathbf{r}$ is independent of the path C joining any two points, then show that there exists a function ϕ such that $\mathbf{V} = \nabla\phi$. (6)

(2)

4. (a) If $\mathbf{V} = y\mathbf{i} + (x - 2xz)\mathbf{j} - xy\mathbf{k}$, then evaluate $\iint_S (\nabla \times \mathbf{V}) \cdot \mathbf{n} dS$, where S is the surface of the sphere $x^2 + y^2 + z^2 = 1$ above the xy -plane. (6)
- (b) Using Green's theorem, show that the area bounded by a simple closed curve C is

$$\frac{1}{2} \int_C (x dy - y dx).$$

Hence obtain the area enclosed by the ellipse $x = 2 \cos \theta$, $y = 3 \sin \theta$. (3+3)

5. (a) Prove Gauss' divergence theorem and show that for a closed surface S

$$\iint_S \mathbf{r} \cdot \mathbf{n} dS = 3V,$$

where V is the volume enclosed by S . (6)

- (b) Verify Stokes' theorem for $\mathbf{V} = (2x - y)\mathbf{i} - yz^2\mathbf{j} - y^2z\mathbf{k}$, where S is the upper half hemisphere of the sphere $x^2 + y^2 + z^2 = 1$ and C is its boundary. (6)
6. (a) Evaluate

$$\iiint_V (x^2 + y^2 + z^2) dx dy dz,$$

where V is the region enclosed by the sphere $x^2 + y^2 + z^2 = a^2$. (6)

- (b) Express $\nabla \times \mathbf{V}$ and $\nabla^2 \phi$ in spherical coordinates in $\mathbb{R}^3$. (3+3)
7. (a) Give definition of Christoffel symbols of second kind and obtain transformation laws for them. (2+4)
- (b) Prove that the covariant derivatives of g_{ij} , g^{ij} , and δ_j^i are zero. (2+2+2)

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